Deep Learning for Precipitation Nowcasting: A Benchmark and A New Model

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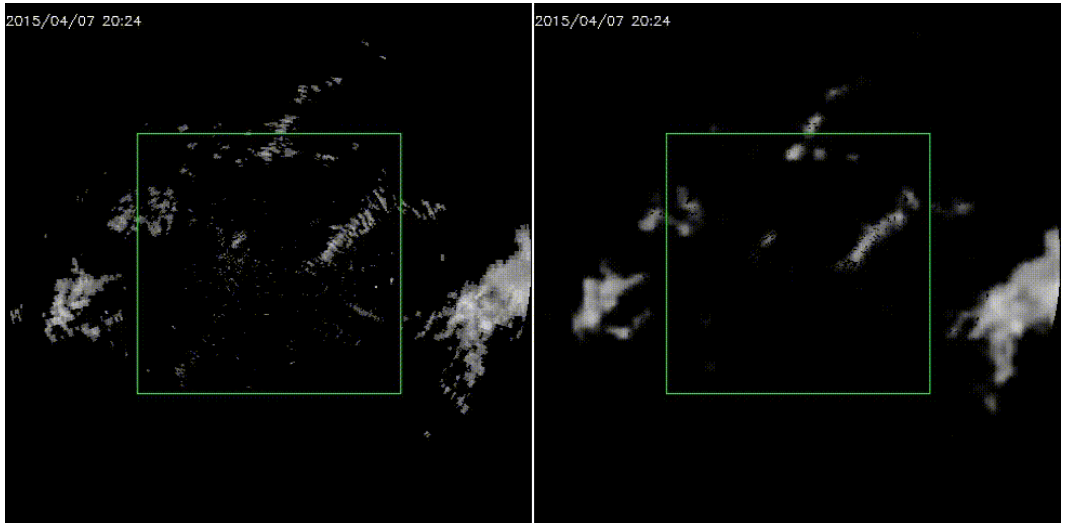
简介

- 降水临近预报是指利用**雷达回波图1、雨量计**等观测数据，结合数值天气预报(NWP)模型，对局部地区降水强度进行极短时间(如0-6小时)预报的问题。



相同点：外形运动特征相似，雨量计（烟雾浓度）
不同点：降水临近预报　　时空序列预测问题
　　烟雾分割　　视频语义分割

模型结构

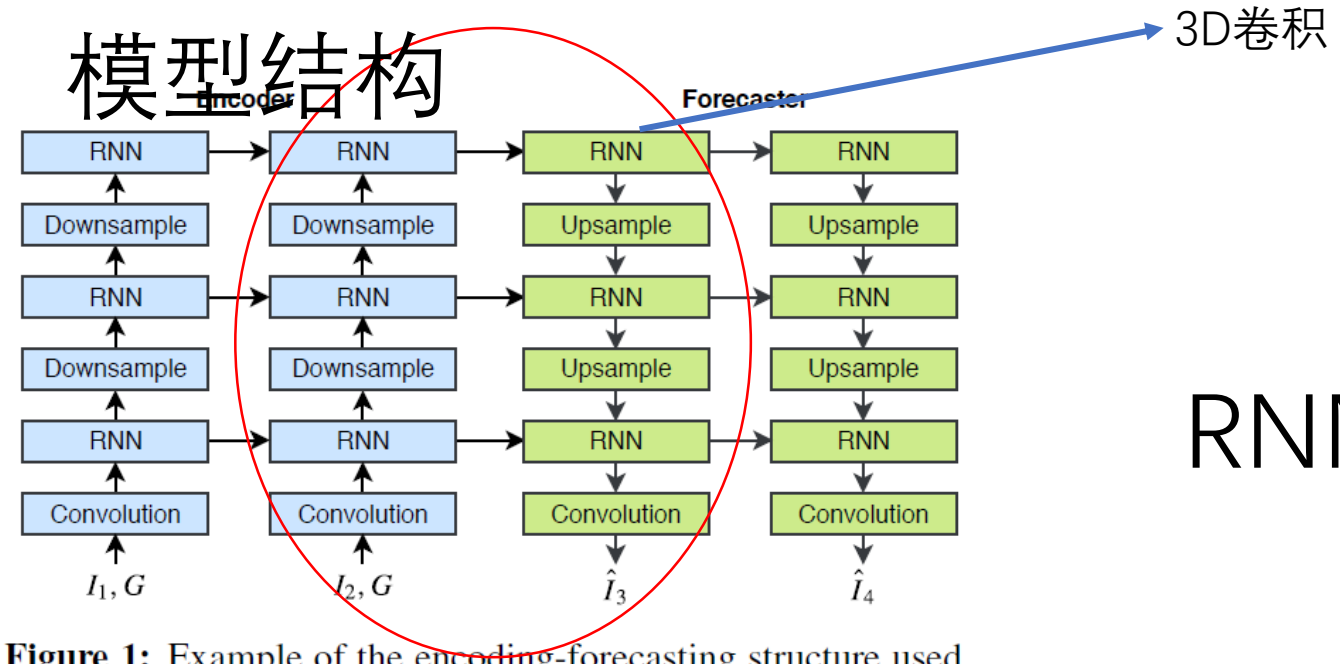


Figure 1: Example of the encoding-forecasting structure used in the paper. In the figure, we use three RNNs to predict two future frames $\hat{I}_3, \hat{I}_4$ given the two input frames I_1, I_2. The spatial coordinates G are concatenated to the input frame to ensure the network knows the observations are from different locations. The RNNs can be either ConvGRU or TrajGRU. Zeros are fed as input to the RNN if the input link is missing.

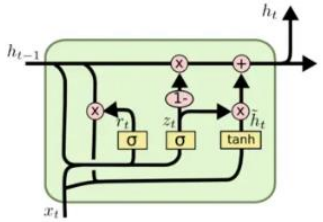
RNN：convgru

trajgru（本文提出）

The reason to reverse the order of the forecasting network is that the high-level states, which have captured the global spatiotemporal representation,could guide the update of the low-level states. Moreover, the low-level states could further influence the prediction.

convGRU

$$\mathcal{Z}_t = \sigma(\mathcal{W}_{xz} * \mathcal{X}_t + \mathcal{W}_{hz} * \mathcal{H}_{t-1}),$$
$$\mathcal{R}_t = \sigma(\mathcal{W}_{xr} * \mathcal{X}_t + \mathcal{W}_{hr} * \mathcal{H}_{t-1}),$$
$$\mathcal{H}'_t = f(\mathcal{W}_{xh} * \mathcal{X}_t + \mathcal{R}_t \circ (\mathcal{W}_{hh} * \mathcal{H}_{t-1})),$$
$$\mathcal{H}_t = (1 - \mathcal{Z}_t) \circ \mathcal{H}'_t + \mathcal{Z}_t \circ \mathcal{H}_{t-1}.$$



$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$
$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$
$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

reset gate r_t 、update gate z_t 、候选隐藏层 $\tilde{h}_t$ （与 LSTM 的 $\tilde{c}_t$ 类似，可以看成是当前时刻的新信息，其中 r_t 用来控制需要保留多少之前的记忆）

为了克服RNN 无法很好处理远距离依赖而提出了 LSTM，而 GRU 是LSTM 的一个变体，GRU 保持了 LSTM 的效果同时又使结构更加简单。
GRU 只剩下两个门，即**更新门**和**重置门**。更新门用于控制前一时刻的状态信息被代入到当前状态的程，更新门的值越大说明前一时刻的状态信息带入越多。重置门用于控制忽略前一时刻的状态信息的程度，重置门的值越小说明忽略得越多。

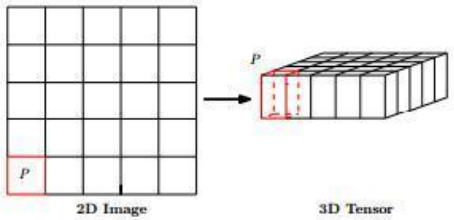


Figure 1: Transforming 2D image into 3D tensor

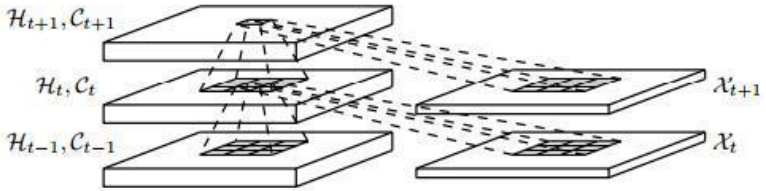


Figure 2: Inner structure of ConvLSTM

However, the convolutional recurrence structure in ConvLSTM-based models is location-invariant while natural motion and transformation (e.g., rotation) are location-variant in general.
Trajectory GRU (TrajGRU) model that can actively learn the location-variant structure for recurrent connections

Trajectory GRU

- 当用于获取时空相关性时，ConvGRU或者其他ConvRNNs等的不足在于所有位置的连接结构和权值都是固定的。卷积运算基本上是对输入应用一个位置不变的滤波器。如果输入都是0，重置门都是1，我们可以在一个特定的位置(i，j) t时刻，重写获取新状态的计算过程，具体如下:

$$\mathcal{H}'_{t,:,i,j} = f(\mathbf{W}_{hh}\text{concat}(\langle \mathcal{H}_{t-1,:,p,q} \mid (p,q) \in \mathcal{N}^h_{i,j} \rangle)) = f(\sum_{l=1}^{|\mathcal{N}^h_{i,j}|} \mathbf{W}^l_{hh}\mathcal{H}_{t-1,:,p_{l,i,j},q_{l,i,j}}). \quad (2)$$

Here, $\mathcal{N}^h_{i,j}$ is the ordered neighborhood set at location (i,j) defined by the hyperparameters of the state-to-state convolution such as kernel size, dilation and padding [30]. $(p_{l,i,j}, q_{l,i,j})$ is the lth neighborhood location of position (i,j). The concat$(\cdot)$ function concatenates the inner vectors in the set and $\mathbf{W}_{hh}$ is the matrix representation of the state-to-state convolution weights.

- 当卷积的超参数固定时，邻域集合N对所有位置保持不变。然而，大多数运动模式对于不同的位置有不同的邻域集。例如，旋转和缩放会产生指向不同方向的不同角度的流场。因此，更合理的做法是使用基于位置变化的连接结构location-variant connection structure：

$$\mathcal{H}'_{t,:,i,j} = f(\sum_{l=1}^{L} \mathbf{W}^l_{hh}\mathcal{H}_{t-1,:,p_{l,i,j}(\theta),q_{l,i,j}(\theta)}), \quad (3)$$

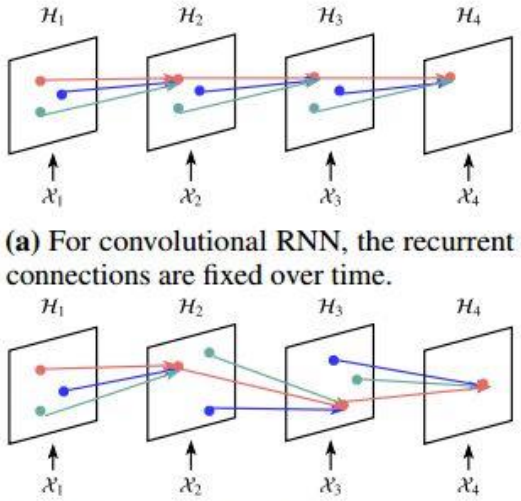
where L is the total number of local links, $(p_{l,i,j}(\theta), q_{l,i,j}(\theta))$ is the lth neighborhood parameterized by θ.

- 基于此，我们提出了TrajGRU，它使用当前输入和以前的状态在每个时间戳为每个位置生成本地邻域集。由于位置指数是离散的、不可微的，我们用一组连续的光流来表示它们“指标”。TrajGRU的主要公式如下:

$$
\begin{aligned}
\mathcal{U}_t, \mathcal{V}_t &= \gamma(\mathcal{X}_t, \mathcal{H}_{t-1}), \\
\mathcal{Z}_t &= \sigma(\mathcal{W}_{xz} * \mathcal{X}_t + \sum_{l=1}^{L} \mathcal{W}_{hz}^l * \mathrm{warp}(\mathcal{H}_{t-1}, \mathcal{U}_{t,l}, \mathcal{V}_{t,l})), \\
\mathcal{R}_t &= \sigma(\mathcal{W}_{xr} * \mathcal{X}_t + \sum_{l=1}^{L} \mathcal{W}_{hr}^l * \mathrm{warp}(\mathcal{H}_{t-1}, \mathcal{U}_{t,l}, \mathcal{V}_{t,l})), \\
\mathcal{H}'_t &= f(\mathcal{W}_{xh} * \mathcal{X}_t + \mathcal{R}_t \circ (\sum_{l=1}^{L} \mathcal{W}_{hh}^l * \mathrm{warp}(\mathcal{H}_{t-1}, \mathcal{U}_{t,l}, \mathcal{V}_{t,l}))), \\
\mathcal{H}_t &= (1 - \mathcal{Z}_t) \circ \mathcal{H}'_t + \mathcal{Z}_t \circ \mathcal{H}_{t-1}.
\end{aligned} \tag{4}
$$

Here, L is the total number of allowed links. $\mathcal{U}_t, \mathcal{V}_t \in \mathbb{R}^{L \times H \times W}$ are the flow fields that store the local connection structure generated by the structure generating network γ. The $\mathcal{W}_{hz}^l, \mathcal{W}_{hr}^l, \mathcal{W}_{hh}^l$ are the weights for projecting the channels, which are implemented by 1×1 convolutions. The $\mathrm{warp}(\mathcal{H}_{t-1}, \mathcal{U}_{t,l}, \mathcal{V}_{t,l})$ function selects the positions pointed out by $\mathcal{U}_{t,l}, \mathcal{V}_{t,l}$ from $\mathcal{H}_{t-1}$ via the bilinear sampling kernel [10, 9]. If we denote $\mathcal{M} = \mathrm{warp}(\mathcal{I}, \mathbf{U}, \mathbf{V})$ where $\mathcal{M}, \mathcal{I} \in \mathbb{R}^{C \times H \times W}$ and $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{H \times W}$, we have:

$$
\mathcal{M}_{c,i,j} = \sum_{m=1}^{H} \sum_{n=1}^{W} \mathcal{I}_{c,m,n} \max(0, 1 - |i + \mathbf{V}_{i,j} - m|) \max(0, 1 - |j + \mathbf{U}_{i,j} - n|). \tag{5}
$$



(a) For convolutional RNN, the recurrent connections are fixed over time.

(b) For trajectory RNN, the recurrent connections are dynamically determined.

Figure 2: Comparison of the connection structures of convolutional RNN and trajectory RNN. Links with the same color share the same transition weights. (Best viewed in color)

这种结构的优点是,我们可以学习子网的连接拓扑通过学习参数γ。在我们的实验中,γ需要连接的Xt和Ht-1作为输入和固定是one-hidden-layer的卷积神经网络（5×5的kernel size和32个filter）。因此,γ只有少量的参数并增加几乎没有成本的整体计算量。